

- 7 Boron trichloride, BCl_3 , and selenium hexafluoride, SeF_6 , both have simple molecular structures.

Which row correctly describes the overall polarity of these two molecules?

	boron trichloride	selenium hexafluoride
A	non-polar	non-polar
B	non-polar	polar
C	polar	non-polar
D	polar	polar

- 8 In a bromine molecule, Br_2 , the two bromine atoms are held together by a single covalent bond.

The bond energy is 193 kJ mol^{-1} . The bond length is $2.28 \times 10^{-10} \text{ m}$.

What do the terms *bond energy* and *bond length* mean?

	bond energy	bond length
A	the energy needed to break one mole of bonds	the distance between the nuclei of the two bonded atoms
B	the energy needed to	the distance between the outer